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PERFORMANCE SPECIFICATION
LASER INFRARED OBSERVATION SET, AN/PVS-6

This performance specification is approved for use by USACECOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This performance specification covers the Laser Infrared Observation Set, referred to herein as the Miniature Eyesafe Laser Infrared Observation Set (MELIOS).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5860

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SPECIFICATIONS

DEPARTMENT OF DEFENSE

*EL-CL-0192-001A APPENDIX A - Nuclear survivability.

*Classified document

HANDBOOKS

DEPARTMENT OF DEFENSE

MIL-HDBK-781 - Reliability Test Methods, Plans and Environments for Engineering Development, Qualification and Production

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia , PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATIONS

DEPARTMENT OF DEFENSE

TITLE 10	- United States Code of Federal Regulations Energy.
TB-MED-524	- Control of Hazards to Health from Laser Radiation.
TM 11-5860-202-10	- Operations Manual, MELIOS, AN/PVS-6

DRAWINGS

USA CECOM

A3187100	- Mini Eyesafe Laser Infrared Observation Set.
A3187116	- Reticle
A3187120	- Laser Rangefinder, MELIOS (AN/PVS-6)
A3187229	- Case, Laser Rangefinder

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A3187238	- Tripod, Assembly
A3187239	- Shipping and Storage Container Assembly MELIOS (AN/PVS-6)

(Copies of other Government documents, drawings, and publications required by suppliers in performance of specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Where specified in the contract (see 6.2), the contractor shall furnish first article units for inspection (see 4.3).

3.2 Materials. The contractor shall select materials that are fully capable of meeting all of the operational and environmental requirements specified herein. The materials specified in the applicable drawing are recommended, but are not mandatory. Selection criteria of the class, grade or type part shall be that the material will be able to perform its intended function when it is assembled. Verification of the supplier meeting the overall performance requirements shall be the governing acceptance standard. Recovered materials shall be used to the maximum extent possible.

3.3 Description. The MELIOS Laser Rangefinder herein identified as the MLRF is a hand held, self contained, laser ranging system. The MLRF consists of the Laser Rangefinder, MELIOS less all ancillary equipment listed in 3.7.13.

3.4 Physical characteristics.

3.4.1 Weight.

3.4.1.1 MLRF weight. The weight of the MLRF excluding battery, shall not exceed 3.7 pounds.

3.4.1.2 MELIOS system weight (fielded). The weight of the MELIOS system, including the MLRF (without battery), tripod, lens cleaning kit, and Laser Rangefinder case, excluding the shipping and storage container, 2 batteries and operator's manual shall not exceed 6.5 pounds.

3.4.1.3 MELIOS system weight (shipping). The weight of the MELIOS system (less batteries), as described in 1.1 shall not exceed 30.0 pounds.

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3.4.2 Finish. Chemical and biological agent resistant materials and finishes shall be used to facilitate decontamination and use of the MLRF in a toxic environment. Finishes shall also be anti-reflective.

3.4.3 Marking. The assembly and all spare components, subassemblies and parts shall be marked in accordance with A3187100.

3.4.4 Workmanship. All optical parts shall be free of material defects such as bubbles, seeds, cracks, feathers, milkiness and other imperfections. Manufacturing defects such as blisters, burns, chips, dirt and other like defects shall be cause for rejection of the component or final assembly.

3.5 Reliability. The MLRF shall provide a Mean Ranges Between Failures (MRBF) of not less than 40,000 ranges. This shall be the lower test (θ_1) MRBF and shall be demonstrated with 80 percent confidence. Replacement of batteries is not included in this requirement.

3.6 Environmental stress screening (ESS). Each MLRF shall accumulate a minimum of 24 hours exposure composed of repetitions of the temperature and power profiles in Figure 2 and the vibration profile per Figure 2 and Figure 5 and with the last two cycles failure free.

3.7 Performance.

3.7.1 Beam characteristics.

3.7.1.1 Operating wavelength. All laser emissions shall be at 1.54 ± 0.02 microns.

3.7.1.2 Pulse width. Each MLRF shall exhibit pulse widths which fall between 16 and 30 nanoseconds inclusive and each pulse width shall equal the mean value x. ± 3 nanoseconds for the MLRF tested.

3.7.1.3 Single pulse. The MLRF shall produce only one laser pulse resulting from each range operation. The MLRF shall emit this pulse only when the “FIRE” switch is depressed in accordance with the correct ranging sequence (see 3.12.5).

3.7.1.4 Beam divergence. Not less than 80 percent of the output energy shall be contained within a 1.0 milliradian diameter beam divergence as measured in the far field.

3.7.1.5 Output energy. The output energy of each MLRF unit shall be between 10 and 33 millijoules. The pre ESS mean measurement shall be between 15 and 19 millijoules. The maximum dispersion about the mean value shall not exceed ± 25 percent of the mean value. The rate of exposure at the output aperture shall not exceed 1 J/CM^2.

3.7.2 Ranging.

3.7.2.1 Range capability. The MLRF shall determine and display in the eyepiece the distance to a target from a minimum of 50 meters to a maximum of 9995 meters in 5 meter increments.

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Attempted rangings at targets which are at a distance of 35 meters or less and targets which are at a distance of 10,005 meters or greater shall result in a display of four zeros (0000) in the eyepiece. The MLRF shall exhibit no more than one failure out of any 100 consecutive rangings thereby determining the range to targets under the conditions that the targets shall be normal to the laser beam, the target size shall not be greater than 10 meters square, the target shall be at a range not less than 9990 meters, the target shall have diffused Lambertian reflectance of not greater than 10 percent at the laser operating wavelength, and the visual range shall be 20,000 meters.

3.7.2.2 Ranging logic. Ranging logic shall be first/last pulse, switch selectable.

3.7.2.3 Target discrimination. The MLRF shall discriminate between targets along the same line-of-sight where range separation is 50 meters or greater and shall provide positive, visual indication for multiple targets.

3.7.2.4 Range error. The range error shall not exceed ± 5 meters.

3.7.2.5 False alarm rate. The false alarm rate shall not exceed one false range reading in one hundred (100) range operations.

3.7.3 Duty cycles.

3.7.3.1 Sustained duty cycle. The MLRF shall be capable of continuous operation at a minimum rate of 6 rangings per minute.

3.7.3.2 Burst rate duty cycle. The MLRF shall be capable of continuously performing at a burst rate duty cycle consisting of 10 rangings per minute for 2 minutes followed by a two minute cool down period.

3.7.4 System timing.

3.7.4.1 Charge time. The MLRF shall provide a positive visual ready-to-fire indication within 3 seconds of activation of the charge switch. The ready-to-fire status of the MLRF, to include the visual indicator in the eyepiece, shall be maintained for 8 ± 1 seconds, after which the unit shall shut down automatically.

3.7.4.2 Display time. The range shall be displayed within 1 second after activation of the fire switch. When the fire switch is released the range shall be automatically displayed for a period of 5 ± 1 seconds, after which the unit shall shut down automatically. Any time the charge switch is activated, the MLRF shall immediately cease to display the range and return to a “ready” status within 3 seconds. The range shall remain displayed when the fire switch is activated and held in the fire position.

3.7.5 System response. The MLRF shall be capable of performing a single ranging operation within 5 seconds after turning the system on.

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3.7.6 System recycling. System recycling shall be automatic between ranging sequences. The battery drain between ranging sequences shall be less than 1 milliampere.

3.7.7 Battery operation.

3.7.7.1 Battery compatibility. The MLRF shall be battery powered and shall use as its primary power source the BA-6516/U Lithium Thionyl Chloride battery. The MLRF shall be capable of utilizing, as an alternate source, the BB-516/U Nickel Cadmium battery. Both batteries shall be field replaceable by the operator without the use of tools (see 6.2).

3.7.7.2 Energy consumed per ranging operation. The MLRF shall consume less than 1200 milliampere seconds per ranging operation at an ambient temperature of 23 ± 5°C (72 ± 9°F). The ranging operation shall include charge time, fire time and display time.

3.7.7.3 Low power indicator. A positive visual indication through the eyepiece showing a low battery condition shall be provided to the operator. When the voltage is monitored at the battery terminals the indicator shall illuminate when the voltage drops to between 20.5 and 19.5 volts dc, (Vdc), and remain illuminated until the shutdown voltage is reached. Alternately, the low power indication shall illuminate when the voltage drops between 21.5 and 20.5 Vdc when power is applied to the test connector, due to the voltage drop of the internal protection diode.

3.7.7.4 Low power activation. The MLRF shall neither be damaged nor suffer subsequent degradation of performance when attempting to operate below the power conditions specified in 3.7.7.3. Inability to obtain a ready signal under these conditions shall not constitute a failure.

3.7.8 Sighting optics.

3.7.8.1 Magnification. Sighting optics magnification shall be not less than 7 power.

3.7.8.2 Field of view. The horizontal field-of-view of the sighting optics shall be 7 degrees minimum. The vertical field-of-view shall be 5 degrees minimum.

3.7.8.3 Visible light transmission. Light transmission of the sighting optics in the visible light range shall be a minimum of 25 percent. This requirement shall be met, in its entirety, with respect to the requirement of the classified document EL-CP-0192-001A Appendix A.

3.7.8.4 Resolution. The sighting optics of the MLRF shall be capable of resolving an object that subtends an angle of not greater than 20 arc seconds.

3.7.8.5 Reticle. The reticle shall be in accordance with A3187116 or similar design. The one (1) milliradian aiming circle shall be boresighted to the transmitter and receiver.

3.7.8.6 Reticle illumination. The MLRF reticle shall have the ability to be illuminated for night and low light level operation. The reticle when not illuminated shall not cause a drain on the

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battery. The reticle illumination shall be actuated by a push/pull switch with intensity varied by rotation of the switch.

3.7.8.7 Optical improvements. The MLRF shall meet the requirements of 30.2.1, 30.2.2 and 30.2.3 included in the classified document EL-CP-0192-0001A Appendix A.

3.7.8.8 Display/visual indicators. All visual indicators shall be positioned in a manner such that, they can be viewed easily through the eyepiece while not reducing the vertical field-of-view to less than requirements of 3.7.8.2. All visual indicators shall be evenly illuminated and of a color and intensity such that when activated they are clearly visible in bright sunlight.

3.7.9 Eyepiece.

3.7.9.1 Diopter. The eyepiece shall provide a diopter adjustment over the range of +4 to -4.

3.7.9.2 Exit pupil. The eyepiece shall have an exit pupil diameter of not less than 5 millimeters.

3.7.9.3 Eye relief. The eyepiece shall have an eye relief of not less than 25 millimeters.

3.7.9.4 Eye shield. The eyepiece shall be equipped with a flexible eye shield which shall reduce extraneous light. The MLRF with eye shield shall be fully functional with all military issue eyeglasses or equivalent.

3.7.10 Flashlamp stray emission. All flashlamp emissions shall be contained within the equipment housing.

3.7.11 Mounting interface alignment. The keyed mounting interface surface shall optically align with the MLRF sighting optics to within 0.25 milliradians and shall not shift more than 0.1 milliradians under any conditions.

3.7.12 Boresight.

3.7.12.1 Boresight error. The boresight error between the Laser Beam and the optical axis of the receiver/sighting telescope shall not exceed 0.5 milliradians.

3.7.13 Ancillary equipment.

3.7.13.1 Tripod. The tripod shall be in accordance with A3187238 or similar design. The mounted MLRF shall remain fixed to the tripod during ranging.

3.7.13.2 Laser rangefinder case. The Laser Rangefinder case shall be in accordance with A3187229 or similar design and shall contain the MLRF tripod, lens cleaning kit, operator's manual, and space for two BA-6516/U batteries. All items shall fit within the closed Laser rangefinder case.

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3.7.13.3 Shipping and storage container. The shipping and storage container shall be in accordance with A3187239 or similar design and shall contain the Laser Rangefinder case packed with the MLRF, tripod, lens cleaning kit, operator's manual, and two BA-6516/U batteries. The shipping case shall be waterproof and protect its contents from shock and vibration while being transported as loose cargo.

3.7.13.4 Lens cleaning kit. The cleaning kit consists of one package of lens cleaning tissue and a 1/2 ounce plastic dropper bottle of lens cleaning fluid. The MLRF shall not be damaged or degraded by use of this lens cleaning kit.

3.8 Environmental requirements.

3.8.1 Temperature.

3.8.1.1 High temperature. The MLRF shall exhibit neither damage (see 3.12.1) nor degradation of performance (see 3.12.2) when operated in temperatures to +49°C (+120°F). The MLRF shall exhibit neither damage, nor degradation of performance after being stored in a non-operating mode in temperature to +71°C (+160°F).

3.8.1.2 Low temperature. The MLRF shall exhibit neither damage nor degradation of performance when operated in temperatures as low as -32°C (-25°F). The MRLF shall exhibit neither damage, nor degradation of performance after being stored in a non-operating mode in temperature as low as -57°C (-70°F).

3.8.1.3 Temperature shock. The MLRF shall exhibit neither damage nor degradation of performance after being subjected, in the non-operating mode, to sudden temperature changes between -57°C (-70°F) and +71°C (+160°F) inclusive.

3.8.1.4 Solar radiation. The MLRF shall exhibit neither damage nor degradation of performance when subjected to temperature fluctuations from +32°C (+89°F) to +49°C (+120°F) with exposure to solar radiation.

3.8.2 Immersion. The MLRF, in the non-operating mode, shall not leak when immersed under 1 meter (3.3 feet) of water for 120 minutes, and upon retrieval shall exhibit neither damage nor degradation of performance during subsequent operation.

3.8.3 Humidity. The MLRF shall not experience any leakage, damage, or degradation of performance resulting from being subjected to a relative humidity and temperature excursion using the recommended parameters and limits per 6.5.6.

3.8.4 Altitude.

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3.8.4.1 Storage and operation. The MLRF shall exhibit neither damage nor degradation of performance due to storage or operation within altitudes up to 15,000 feet in an unprotected operating mode.

3.8.4.2 Rapid decompression. The MLRF, in the Laser rangefinder case, shall not present a safety or health hazard to personnel, or aircraft either during, or after exposure to an equivalent altitude change from 15,000 to 40,000 feet, which occurs within 15 seconds. Additionally, the MLRF shall present neither a safety nor health when subsequent operation is attempted.

3.8.5 Fungus. The MLRF, Laser Rangefinder case, tripod and shipping and storage container shall not support fungus growth, nor experience damage, nor degradation of performance by the presence of fungus spores or adjacent fungus growth.

3.8.6 Vibration.

3.8.6.1 Field vibration. The MLRF shall exhibit neither damage nor degradation of performance by exposure to a thirty (30) minute vibration, 10 minutes in each orthogonal axis.

3.8.6.2 Transportation vibration. The MLRF, in its shipping and storage container, containing all the ancillary items, shall exhibit neither damage, nor degradation of performance after being subjected to vibrations typically induced when transported as loose cargo hostile field conditions.

3.8.7 Shock.

3.8.7.1 Field shock. The MLRF shall exhibit neither damage nor degradation of performance resulting from physical shocks typically encountered during handling, and field use.

3.8.7.2 Transportation shock. When the MLRF is packaged in its shipping and storage container with all the ancillary items, it shall not be damaged, or exhibit degradation of performance as a result of physical shocks incurred during typical field transportation, including loading and unloading from tracked and wheeled vehicles, and manportable transportation.

3.8.7.3 Bench handling. The MLRF shall exhibit neither damage nor degradation of performance after experiencing mechanical shocks commonly induced during repair, maintenance, and system checkout.

3.8.8 Sand and dust. The MLRF shall exhibit neither damage nor degradation of performance by exposure, with lens covers installed, to blowing sand and dust..

3.8.9 Salt fog. The MLRF and the shipping and storage container shall exhibit neither damage nor degradation of performance due to exposure to a salt atmosphere.

3.8.10 Explosive atmosphere. The MLRF shall not initiate combustion when operated in an atmosphere that is heavily laden with fumes of aircraft or ground vehicle fuels.

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3.9 Electromagnetic interference (EMI). The MLRF shall comply with the radiated emission and susceptibility requirements in Appendix A.

3.10 Nuclear survivability. The MLRF including all mission-essential mechanical configurations, optical components, electronic assemblages, electronic equipment, electronic circuits and electronic piece parts shall withstand the nuclear environments as specified in the classified document EL-CP-0192-0001A Appendix A, paragraph 30.1. The MLRF shall meet the operational requirements of section 3.7.2, of this specification after being subjected to the specified nuclear environments. The following parameters are specified criteria in Figure A-10 and A-11, and Table A-3, which are included in the Classified Document EL-CP-0192-0001A Appendix A.

a. Time dependent total electric field (E) in volts/meter.

b. Time dependent total magnetic field (H) in amperes/meter.

c. The frequency content of the electric field (E(f)) in volts/meter/hertz.

d. Polarization.

3.11 Safety. The MLRF laser output shall be measured and the MLRF shall be classified and marked in accordance with TB-MED-524, Control of Hazards to Health from Laser Radiation dated 30 May 1975. Potentially hazardous areas within the MLRF shall be identified (see 6.6). Optical products shall contain no thorium or other source material as defined by Title 10, Code of Federal Regulations Energy, Part 40, in excess of 0.05 percent by weight (500 ppm) or any other added radioactive materials.

3.12 Technical interpretations. The following technical interpretations are, when referenced in sections 3, 4, or 5, mandatory for this specification.

3.12.1 Damage. Damage is the deterioration of any physical characteristics including, but not limited to , surface material, component, durability and safety or the failure of coatings, adhesives, materials, hardware and fasteners.

3.12.2 Degradation of performance. Degradation of performance is a cumulative degradation of a measured level of performance while remaining within the specified performance requirements. For reliability tests degradation of performance is the same as the limits of the specified performance requirements. For all other tests or series of tests, degradation of performance is the failure of any MLRF to perform within the specified requirements as well as the following criteria.

a. The post-ESS mean pulse width shall remain within +3/-5 nanoseconds of the pre-ESS mean pulse width value, and for all test results subsequent to post-ESS, the mean pulse width shall remain within ± 3 nanoseconds of the post-ESS mean value.

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b. The post-ESS mean output energy shall remain within +3/-2 millijoules of the pre-ESS mean output energy shall remain within ± 3 millijoules of the post-ESS mean value.

c. The beam divergence measurement shall conform to the requirements of 3.7.1.4.

d. The Boresight shift shall conform to the requirements of 3.7.12.2.

3.12.3 Maximum lot size. The maximum allowable inspection lot after IPT and the first production lot shall be not greater than 300 units.

3.12.4 Successful range operation. A successful range operation is where the MLRF is operated through a ranging sequence (see 3.12.5) and displays a range reading that is within ± 5 meters of the actual target distance.

3.12.5 Ranging sequence. A ranging sequence is defined as follows:

a. Select First or Last RTN position.

b. Press the “Charge” button.

c. Wait for the “Ready” light to illuminate.

d. Press the “Fire” button.

e. Read the displayed range.

4. VERIFICATION

4.1 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.2 Inspection conditions.

4.2.1 Inspection power source. Unless specifically specified all inspection/tests specified herein, shall use the BA-6516/U or BB-516/U battery (or commercial equivalent) or external 24 Vdc power supply as the MLRF power source.

4.2.2 Sequence of inspections. Unless otherwise, specified herein inspections and tests shall be performed in any order.

4.2.3 Testing sequences. Upon successful completion of ESS all First Article units shall be subjected to the inspections of Table II, Subgroup 1. When Subgroup 1 inspections are completed

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Subgroups 2, 3 and 4 shall be performed. A minimum of: Six (6) units shall be assigned to Subgroup 2, Three (3) units assigned to Subgroup 3, and Six (6) units assigned to Subgroup 4 (reliability). For the inspections of Subgroup 2 only three (3) units are required to be subjected to, Temperature Shock, Vibration, Shock and Immersion in this order. For the inspections of Subgroup 3 marked (X91) only one unit is required to be subjected to any one test. Except for these specific sequences, all other tests may be performed in any order.

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TABLE II. First article inspection.

Requirement	Requirement paragraph	Verification paragraph	Subgroup 1 (15) (min)	2 (6) (min)	3 (3) (min)	4 (6) (min)
Weight	3.4.1	Measured			X	
Operating wavelength	3.7.1.1	4.7.1.1			X	
Pulse width	3.7.1.2	4.7.1.2	X			
Single pulse	3.7.1.3	4.7.1.3	X			
Beam divergence	3.7.1.4	4.7.1.4	X			
Output energy	3.7.1.5	4.7.1.5	X			
Range capability	3.7.2.1	4.7.2.1	X			
Ranging logic	3.7.2.2	4.7.2.2	X			
Target discrimination	3.7.2.3	4.7.2.3	X			
Range error	3.7.2.4	4.7.2.4	X			
False alarm rate	3.7.2.5	4.7.2.5	X			
Sustained duty cycle	3.7.3.1	4.7.3.1	X			
Burst rate duty cycle	3.7.3.2	4.7.3.2	X			
Charge time	3.7.4.1	4.7.4.1		X		
Display time	3.7.4.2	4.7.4.2		X		
System response	3.7.5	4.7.5		X		
System recycling	3.7.6	4.7.6		X		
Battery compatibility	3.7.7.1	4.7.7.1		X		
Energy consumed per ranging	3.7.7.2	4.7.7.2		X		
Low power indicator	3.7.7.3	4.7.7.3		X		
Low power activation	3.7.7.4	4.7.7.4		X		
Magnification	3.7.8.1	4.7.8.1			X	
Field of view	3.7.8.2	4.7.8.2			X	
Visible light transmission	3.7.8.3	4.7.8.3			X	
Resolution	3.7.8.4	4.7.8.4			X	
Reticle	3.7.8.5	4.7.8.5			X	
Reticle illumination	3.7.8.6	4.7.8.6			X	
Optical improvement	3.7.8.7	4.7.8.7			X	
Diopter	3.7.9.1	4.7.9.1			X	
Display/visual indicators	3.7.8.8	4.7.8.8			X	
Tripod	3.7.13.1	4.7.13.1			X	
Shipping and storage container	3.7.13.3	4.7.13.3			X(1)	

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TABLE II. First article inspection. - Continued

Requirement	Requirement paragraph	Verification paragraph	Subgroup 1 (15) (min)	2 (6) (min)	3 (3) (min)	4 (6) (min)
Laser rangefinder case	3.7.13.2	4.7.13.2			X(1)	
Lens cleaning kit	3.7.13.4	4.7.13.4			X	
Exit pupil	3.7.13.2	4.7.13.2			X	
Eye relief	3.7.13.3	4.7.13.3			X	
Eye shield	3.7.9.4	4.7.9.4			X	
Flashlamp stray emission	3.7.10	4.7.10		X		
Mounting interface alignment	3.7.11	4.7.11		X		
Boresight error	3.7.12.1	4.7.12.1	X			
Boresight shift	3.7.12.2	4.7.12.2	X			
Reliability	3.5	4.5				X
High temperature-storage	3.8.1.1	4.8.1.1		X		
High temperature operation	3.8.1.1	4.8.1.1		X		
Low temperature	3.8.1.2	4.8.1.2		X		
Temperature shock	3.8.1.3	4.8.1.3		X		
Solar radiation	3.8.1.4	4.8.1.4		X		
Immersion	3.8.2	4.8.2		X		
Humidity	3.8.3	4.8.3		X		
Storage and operation	3.8.4.1	4.8.4.1			X	
Rapid decompression	3.8.4.2	4.8.4.2			X(1)	
Fungus	3.8.5	4.8.5			X(1)	
Vibration	3.8.6	4.8.6		X		
Shock	3.8.7	4.8.7		X		
Sand and dust	3.8.8	4.8.8			X(1)	
Salt fog	3.8.9	4.8.9			X(1)	
Explosive atmosphere	3.8.10	4.8.10			X(1)	
Electromagnetic interference	3.9	4.9			X(1)	
Nuclear survivability	3.10	4.10			X(1)	
Safety	3.11	4.11			X(1)	

4.4 Conformance inspection. Inspection shall be as specified in Table I and shall be conducted on all MELIOS systems.

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4.4.1 Group A inspection. Group A inspections listed in Table III shall be conducted on all MLRF units which have passed the inspections of Table I and ESS specified in 4 inclusive. Failure of any test shall constitute cause for rejection of that unit. Units successfully meeting the requirements of Group A inspection shall be formed into lots (see 6.2.c) for the purposes of B, C and D inspection.

TABLE III. Group A inspection.

Requirement	Requirement paragraph	Verification paragraph
Pulse width	3.7.1.2	4.7.1.2
Single pulse	3.7.1.3	4.7.1.3
Beam divergence	3.7.1.4	4.7.1.4
Output energy	3.7.1.5	4.7.1.5
Range capability	3.7.2.1	4.7.2.1
Ranging logic	3.7.2.2	4.7.2.2
Target discrimination	3.7.2.3	4.7.2.3
Range error	3.7.2.4	4.7.2.4
False alarm rate	3.7.2.5	4.7.2.5
Sustained duty cycle	3.7.3.1	4.7.3.1
Burst rate duty cycle	3.7.3.2	4.7.3.2
Boresight error	3.7.12.1	4.7.12.1
Boresight shift	3.7.12.2	4.7.12.2

4.4.2 Group B inspection. Group B inspection shall be conducted on samples selected from lots which have passed the tests of 4.4.1. Sampling plans shall be as specified in the contract or purchase order (see 6.2). Group B tests listed in Table IV may be performed in any order.

TABLE IV. Group B inspection.

Requirement	Requirement paragraph	Verification paragraph
Subgroup 1		
Flash lamp stray emission	3.7.10	4.7.10
Charge time	3.7.4.1	4.7.4.1
Display time	3.7.4.2	4.7.4.2
System response	3.7.5	4.7.5
Battery compatibility	3.7.7.1	4.7.7.1
Subgroup 2		
Low power indicator	3.7.7.3	4.7.7.3
Low power activation	3.7.7.4	4.7.7.4
Mounting interface alignment	3.7.11	4.7.11

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4.4.3 Group C inspection. Group C inspection shall be conducted on samples selected from lots which have passed Group A inspection. The sample units shall be tested in accordance with the inspections listed in Table V. Tests may be performed in any order except shock and vibration shall be performed before immersion.

4.4.3.1 Sampling for Group C inspections. Sampling for Group C inspection shall be as specified in the contract or purchase order (see 6.2.f).

4.4.3.2 Group C failures. Actions regarding Group C failures shall be as specified by the contract or purchase order (see 6.2.d).

4.4.3.3 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract or purchase order provided all damage is repaired and the sample units are resubjected to, and pass, Group A inspection.

TABLE V. Group C inspection.

Requirement	Requirement paragraph	Verification paragraph
Subgroup 1		
Field shock	3.8.7.1	4.8.7.1
Field vibration	3.8.6.1	4.8.6.1
Immersion	3.8.2	4.8.2
Subgroup 2		
High temperature-operation	3.8.1.1	4.8.1.1.2
Low temperature proc. I and II	3.8.1.2	4.8.1.2
Temperature shock	3.8.1.3	4.8.1.3
Subgroup 3		
Humidity	3.8.3	4.8.3
Subgroup 4		
Storage and operation	3.8.4.1	4.8.4.1

4.4.4 Group D inspection. Group D inspection shall be performed on samples selected from units which have passed Group A. Testing shall be conducted in accordance with Table VI.

4.4.4.1 Sampling for Group D inspection. The reliability acceptance test shall be performed on each inspection lot. Six units shall be randomly selected from each lot.

4.4.4.2 Group D failures. Actions regarding Group D failures shall be as specified in the contract or purchase order (see 6.2.d).

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4.4.4.3 Re-inspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and passed Group D inspection may be accepted on the contract or purchase order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection. As a minimum, the flashlamp, PFN Capacitor, and laser rod shall be replaced.

TABLE VI. Group D inspection.

Requirement	Requirement paragraph	Verification paragraph
Reliability	3.5	4.5

4.5 Reliability. The MLRF including a battery or simulated battery mass, shall be tested to meet the requirements of 3.5. Battery failures shall not be charged against the reliability requirement. The test profiles to be used for this reliability test are shown in Figures 1 and 4. During each period of "on time", the units shall be operated continuously at the sustained duty cycle. The unit shall be continuously monitored for proper operation during the reliability test. While in the chamber, the occurrence of 3 misfires within any one minute period or 7 misfires within a 48 hour period shall constitute a failure. Performance shall be verified by performing Group A tests before and after each 48 hour test period. Upon failure, if the time to failure cannot be determined, only the time to the last successful measurement of the failed parameter shall count towards fulfilling the test plan requirements. The UTT may be powered by a hard power supply but the battery or a simulated battery containing the same size and mass must be in the system during all reliability testing. Guidance for reliability testing may be found in MIL-HDBK-781 using Test Plan 20-6.

4.6 Environmental stress screening (ESS). Prior and subsequent to ESS the MLRF shall have passed Group A inspection, Table III, and the workmanship requirement of 3.4.4. Each MLRF shall be subjected to a minimum 24 hour ESS test period of which the last 12 hours shall be failure free. The cycle in which a failure occurs shall not be counted towards the 24 hour total. The ESS test shall be comprised of repetitions of the test cycle identified by Figure 2. The unit under test shall be vibrated in accordance with Figure 5 for a minimum of 5 minutes at both hot and cold temperatures in each of the three orthogonal axes. During the periods of applied power the unit under test shall be operated at the sustained duty cycle rate and monitored for successful operation to assure positive detection of any failure at the moment of occurrence. The occurrence of 3 or more misfires detected in the chamber within any one minute period or 7 misfires within a 24 hour period shall constitute a failure. Should a failure occur during the last 12 hours the MLRF shall be reworked and retested to include all required vibration until a minimum of 12 failure free hours have been achieved. Once ESS testing has commenced, no adjustment nor rework shall be made except as required to correct a failure of the unit and return it to test.

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4.7 Methods of inspection.

4.7.1 Beam characteristics.

4.7.1.1 Operating wavelength. Using a narrow band pass filter and a beamsplitter both calibrated at 1.54 microns, and two fast response radiometers verify the MLRF laser output wavelength. Position the MLRF and the beamsplitter in a manner such that, the MLRF laser output strikes the 2 radiometers. Cover the MLRF sighting/receiving optics, fire the MLRF five times and record the reading on both radiometers. Establish that the two radiometers read the same, or that the percent difference is the same for each shot. Place the narrow band pass filter between one of the radiometer detectors and beamsplitter. Cover the MLRF sighting optics. Fire the MLRF five times and record the readings on each radiometer. Compare the readings of the radiometers in the second set of five firings. The energy of the unfiltered beam should be equal to the energy of the filtered beam, allowing for loss through the filters and the difference between the two radiometers. If the readings are not equal, identify the wavelength(s) and intensity(s) of the other beam component(s). As an alternative, a monochrometer may be used to measure the output wavelength of the laser. Scan the wavelength of the laser light emitted from the MLRF from 0.7 to 2.0 microns. The laser radiation received for wavelengths less than 1530 nanometers or greater than 1550 nanometers shall be zero.

4.7.1.2 Pulse width. Position a fast response photodiode sensitive to 1.54 micron radiation in front of the MLRF in order to receive the transmitted pulse from the MLRF. Connect the output of the photodiode to a fast response storage oscilloscope equipped with a scope camera or an equivalent recording device. Retain the photograph or record for use in 4.1.3. Cover the MLRF sighting optics and fire the MLRF five (5) times. Determine the pulse width at half amplitude from the stored image. Record and average the five (5) pulse widths.

4.7.1.3 Single pulse. Using the test results from 4.1.2 examine the results of each repetition for evidence of double or multiple pulsing. Sequence the range operation incorrectly and verify that the MLRF fires only when the fire button is depressed in accordance with the correct ranging sequence (see 3.12.5).

4.7.1.4 Beam divergence. Using an optical bench, a radiometer calibrated for pulsed energy of 1.54 microns, an aperture plate with a 1.0 millirandian diameter hole, and a lens or mirror with sufficient aperture to accommodate the MLRF sighting/receiving optics, measure and record the beam divergence of the MLRF. Center the MLRF reticle on the approximate center of the lens. Align the radiometer to receive the reflected pulse from the mirror. Cover the MLRF sighting/receiving optics. Fire the MLRF five (5) times. Record and average the output energy of the laser. Place the aperture plate at the focal point between the lens/mirror and the radiometer. Position the aperture so that the reticle is centered on the opening in the aperture plate. Fire the MLRF as necessary and adjust the aperture plate for the maximum energy reading. Cover the MLRF sighting optics. Fire the MLRF five (5) more times, record and average the results of output energy. Divide the average of the second five firings by the average of the first five firings. Multiply the result by 100, this yields the percent of energy passing through the 1.0 milliradian aperture.

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4.7.1.5 Output energy. Using a calibrated radiometer, measure the output energy of 10 consecutive shots.

4.7.2 Ranging.

4.7.2.1 Range capability. Using a detector sensitive to 1.54 micron radiation, a delay circuit capable of providing delays corresponding to ranges from 35 meters to 10,005 meters in 5 meter increments and a short pulse light emitting device which emits radiation at the MLRF wavelength, verify the MLRF range performance. The emitter shall radiate energy at the minimum sensitivity of the MLRF receiver. This minimum sensitivity shall be predetermined by ranging to a target as described in 3.7.2 or equivalent and observing the analog output of a MLRF receiver. The test set emitter's output energy shall simulate the return energy from the conditions specified in 3.7.2.1. The test set emitter shall be adjusted to produce three different intensities. (One intensity for near range, one intensity of mid-range and one intensity for long ranges). The test set timing circuit shall be capable of time delays sufficient to simulate ranging from 35 to 10,005 meters, multiple targets, and be selectable to at least 12 specific ranges that exercises all range display segments. Select and record the time delay/range setting. Fire the MLRF at the sustained duty cycle rate. The delay shall be varied to select ranges of 35, 55, 150, 500, 1000, 2500, 5000, 7500, 9000, 9500, 9990, and 10,005 meters. At each range selected, the MLRF shall be fired not less than 10 times and the displayed range shall be the selected range within ± 5 meters and shall be recorded.

4.7.2.2 Ranging logic. The test setup of 4.7.2.1 may be used. It must be capable of providing both a single optical pulse, and two optical pulses separated by 333 nanoseconds (50 meters). In the two pulse mode, the first pulse shall occur at the delay setting of the delay device and the second pulse shall occur 333 nanoseconds later. With the test device in the single pulse mode and the MLRF in the First Target mode, fire the MLRF into the test device. The MLRF shall indicate the proper range within ± 5 meters for the first pulse and the multitarget indicator shall be off. Repeat with test device in the two target mode. The MLRF should display the same range with the multitarget indicator illuminated. Switch the MLRF to the last target mode, and fire the MLRF into the test device. The MLRF should display a range 50 ± 5 meters greater than the range delay selected. The multi-target indicator shall be illuminated. Switch the test device back to the one target mode and keep the MLRF in the last target mode. Fire the MLRF into the test device. The MLRF should display the same range as that selected on the test device and the multitarget indicator should not be illuminated.

4.7.2.3 Target discrimination. Verify the results of the Ranging logic test (see 4.7.2.2).

4.7.2.4 Range error. Verify the results of the Range Capability Test (see 4.7.2.1).

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4.7.2.5 False alarm rate. Using the test set up of 4.7.2.1, set the time delay between laser firing and returned emitter pulse to be equivalent to 15,000 meters. This simulates covering the receiving optics. Fire the MLRF 20 times, at the burst rate duty cycle followed by a two (2) minute cooldown period followed by another 20 shots at burst rate with another 2 minute cool down period, and 60 firings at the sustained duty cycle. Verify the laser fired each time. Record the value of the range for each time the laser fired. The range reading shall be 0000 each time the laser fired.

4.7.3 Duty cycles.

4.7.3.1 Sustained duty cycle. Verify the results of 4.7.2.1. Operation at the specified repetition rate of 6 ranges per minute shall demonstrate the sustained duty cycle.

4.7.3.2 Burst rate duty cycle. Verify the results of 4.7.2.5. Operation at the specified repetition rate of 10 ranges per minute for 2 minutes, with a 2 minute cool down period followed by another 20 shots at burst rate followed by another 2 minute cool down period shall demonstrate the burst rate duty cycle.

4.7.4 System timing.

4.7.4.1 Charge time. Using a time measuring device capable of resolving 0.1 second intervals, measure the time required for the system to indicate a fully charged status. Measure the length of time that the ready-to-fire indicator is maintained before automatic shutdown.

4.7.4.2 Display time. Measure the length of time from activation of the “FIRE” switch to the display of the range. Measure the time that the range continues to be displayed after the “FIRE” switch is released. Perform the ranging operation again and verify that the display remains illuminated while holding the “FIRE” switch down for a period of one minute. Perform the ranging operation a third time and depress the “CHARGE” button within four seconds after the range is displayed and verify that the system returns to a ready-to fire status within 3 seconds.

4.7.5 System response. Using a time measuring device capable of resolving 0.1 second intervals, measure the time from the system turn-on to range display. Turn the MLRF on and begin timing. Press the charge button. When the ready light illuminates, press the FIRE button. When the range is displayed, stop the timer.

4.7.6 System recycling. Initiate the ranging sequence of the MLRF (see 3.12.5). Note and record that the MLRF has ranged and release the buttons depressed for the ranging sequence. Note and record that the range display shuts off within 5 ± 1 seconds. Initiate the ranging sequence again. Note and record that the MLRF ranges a second time. Using the test set up of 4.7.7.2 verify that there is no battery drain between ranging operations in excess of 1 milliampere.

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4.7.7 Battery operation.

4.7.7.1 Battery compatibility. Install a BA-6516/U in the MLRF without the use of tools. Verify that the unit functions properly by performing a successful range operation (see 3.12.4). Repeat using a BB-516/U.

4.7.7.2 Energy consumed per ranging operation. Energize the MLRF from a 24 ± 1 volt external supply through a precision 0.1 ohm current sampling resistor in the return lead, attached to the test connector. Connect a storage oscilloscope across the sampling resistor. Perform a ranging sequence (see 3.12.5). Record the oscilloscope display retaining the photograph or record for use. by dividing the current waveform into more than ten parts, calculate the current and time product of each part from the oscilloscope waveform in milliampere seconds. Add the parts together to derive the total milliampere seconds.

4.7.7.3 Low power indicator. Energize the MLRF through the battery terminals or through the test connector from an external power source adjusted to 22 Vdc applied to the battery terminal or to pin 7 of the test connector with the return connected to pin 3. Decrease the input voltage in 0.2 volt increments, cycling the MLRF through a ranging sequence at each incremental voltage level. Record the voltage level at which the low power indicator (blinking display) is activated.

4.7.7.4 Low power activation. Continue with the test set-up of 4.7.7.3, lower the voltage until the MLRF stops performing. Increase the voltage to 24 Vdc. Perform a ranging operation.

4.7.8 Sighting optics.

4.7.8.1 Magnification. Measure the diameter of the entrance and exit pupil of the MLRF. The ratio of entrance to exit pupil size is the magnification.

4.7.8.2 Field-of-view. Mount the MLRF on a fixture which provides an azimuth readout accurate to the nearest minute. Position the MLRF so that a convenient vertical object is at the left edge of the field of view of the sighting optics. Record the azimuth. Traverse the MLRF until the object is at the right edge of the field of view. Record the azimuth. The difference in azimuth is the horizontal field of view. Reposition the MLRF on its side. Repeat along the vertical axis of the sighting optics. The difference in the readings is the vertical field of view.

4.7.8.3 Visible light transmission. Use a collimated beam of visible light with a limiting aperture. The aperture should be just slightly smaller than the aperture of the MLRF's objective lens. Place an integrating sphere so that all of the light passing through the aperture enters one of its ports. Using the photopic filter measure the intensity of the light in the sphere. Position the MLRF so that light passing the aperture passes through the MLRF and then enters the sphere. Again measure the intensity in the sphere. Calculate transmission by dividing the latter reading by the former.

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4.7.8.4 Resolution. Mount a USAF 1951 resolution target at the focal plane of a collimator. Using a dioptometer, view the target through the MLRF. Calculate the pattern which corresponds to a 20 arc seconds with the focal length of the collimator used. Verify that the sighting optics will resolve this target.

4.7.8.5 Reticle. Visually inspect the MLRF reticle. Failure to meet requirement 3.7.8.5 shall constitute failure of this test.

4.7.8.6 Reticle illumination. Using the test set-up of 4.7.7.2 energize the MLRF through the test connector, charge the system, view the reticle illumination in a dark or low light environment by turning the mode switch to FIRST RTN position and activate the reticle. The reticle shall be visible and readable. Using the current measurement method described in 4.7.7.2, verify that the reticle illumination does not consume battery energy between ranging operations with the reticle not activated.

4.7.8.7 Optical improvement. See 40.1 and 40.2 of the classified Appendix A for conformance requirements.

4.7.8.8 Display/visual indicators. With the MLRF in the test mode activate all the display segments. Verify that all display segments are visible and legible and appear to be equally illuminated. Conduct the test in bright sunlight.

4.7.9 Eyepiece.

4.7.9.1 Diopter. Use a dioptometer or equivalent instrument. Focus the dioptometer to its reticle and se the scale to zero diopter. Place the dioptometer on the eyepiece of the Optics Module. Move the diopter ring of the Optics Module clockwise as far as it will adjust. Refocus the dioptometer on the Optics Module reticle. Read and record the scale on the dioptometer. Move the diopter ring on the Optics Module counter clockwise as far as it will adjust. Again, refocus the dioptometer on the reticle of the Optics Module. Read and record the scale on the dioptometer.

4.7.9.2 Exit pupil. Measure the diameter of the exit pupil, using the reticle in a dynameter K and E Model 71-7105, or equivalent.

4.7.9.3 Eye relief. Without disturbing the setting used for the previous test (Exit Pupil 4.7.9.2), determine the eye relief by reading the focus scale on the side of the dynameter. Calculate the eye relief by adding to this value the distance from the top surface of the eyepiece lens retaining ring to the top surface of the eyepiece lens.

4.7.9.4 Eye shield. Perform a ranging operation with the MLRF using a chemical, biological, and radiological protective mask and perform using standard eye glasses.

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4.7.10 Flashlamp stray emission. Insert an opaque plug in the MLRF eyepiece to block radiation from the range display. Aim the MLRF at a diffuse black surface. Position three observers which have been dark adapted (see 3.12.7) so that they do not view the output window surface of the MLRF nor the reflecting surface at which it is fired. The area in which the test is conducted shall be a darkened room. Each of the three observers shall view the firing of the MLRF at a distance of 5 ± 1 meters with the naked eye and using the AN/PVS-5, Night Vision Goggle or an equivalent.

4.7.11 Mounting Interface Alignment. Align the elements so that the target is on the center line of the collimator and the V mount is aligned parallel to this center line. In addition the "V" mount must be positioned such that a MLRF mounted on it will View the collimated image of the target. Measure the displacement of the target with respect to the MLRF reticle.

4.7.12 Boresight.

4.7.12.1 Boresight error. Align the receiver/sighting optics of the MLRF with the aperture fixture and lens. With the aperture fixture at the focal points of the lens insert a piece of exposed Polaroid film marked with a cross. Align the reticle pattern of the MLRF with the reticle pattern on the film. Cover the MLRF objective lens and fire the MLRF to burn a spot on the film. Measure the distance from the center of the burn spot to the center of the reticle pattern on the film and convert the result to milliradians. Conduct this test a total of (5) five times.

4.7.12.2 Boresight shift. Examine the data of 4.7.12.1 taken before and after each environmental test. Determine and identify the geometric location of the average pretest and posttest boresight error. Compare the data and calculates any difference in boresight between the pretest data and the post test data.

4.7.13 Ancillary equipment inspection.

4.7.13.1 Tripod. Mount the MLRF on the tripod. Inspect to verify that the tripod adjustment remains fixed through at least 5 ranging operations.

4.7.13.2 Laser rangefinder case. Pack the Laser Rangefinder case with all of the items required as identified in 3.7.13.2 and inspect to assure fit of all items. Inspect to assure proper and complete closure of the case.

4.7.13.3 Shipping and storage container. Remove and replace the MLRF and the ancillary equipment from the container. Failure of the shipping container to provide a secure fit, failure of the shipping container.

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4.7.13.4 Lens cleaning kit. Perform the lens cleaning procedure for the MLRF by wetting the cleaning tissue with the solvent and wiping the lenses as described in TM 11-5860-202-10. Inspect to assure that the MLRF has not been damaged or degraded by the cleaning operation and perform a range operation. Evidence of damage or degradation of performance to the MLRF as the result of the cleaning operation shall constitute failure of the MLRF to meet the requirement of 3.7.13.4.

4.8 Environmental tests. For applicable definitions of damage and degradation of performance, see 3.12.1 and 3.12.2. For the purpose of environmental testing, power source failures shall be documented as non-relevant failures. The workmanship inspection of Table I and the Group A inspections listed in Table III shall be performed before and after each individual environmental test. The tests performed after an environmental test may serve as the tests performed before the next environmental test providing all of the following are met:

a. Not more than 72 hours has elapsed since the last tests were completed.

b. The unit under the test has been secured and not tested or operated during that period.

c. The previous tests revealed no failures.

4.8.1.1 High temperature storage. Verify that the high temperature storage requirements of 3.8.1.1 have been met. A recommended method for verification may be found in 6.5.

4.8.1.1.2 High temperature operation. Verify that the high temperature operation requirements of 3.8.1.1.2 have been met. A recommended method for verification may be found in 6.5.

4.8.1.2 Low temperature. Verify that the low temperature requirements of 3.8.1.2 have been met. A recommended method for verification may be found in 6.5.

4.8.1.3 Temperature shock. Verify that the temperature shock requirements of 3.8.1.3 have been met. A recommended method for verification may be found in 6.5.

4.8.1.4 Solar radiation. Verify that the solar radiation requirements of 3.8.1.4 have been met. A recommended method for verification may be found in 6.5.

4.8.2 Immersion. Verify that the immersion requirements of 3.8.2 have been met. A recommended method for verification may be found in 6.5.

4.8.3 Humidity. Verify that the humidity requirements of 3.8.3 have been met. A recommended method of verification may be found in 6.5.

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4.8.4 Altitude.

4.8.4.1 Storage and operation. Verify that the altitude storage and operation requirement have been met. A recommended method for verification may be found in 6.5.

4.8.4.2 Radio decompression. Verify that the rapid decompression requirements of 3.8.4.2 have been met. A recommended method for verification may be found in 6.5.

4.8.4.5 Fungus. Verify that the fungus requirements of 3.8.5 have been met. A recommended method for verification may be found in 6.5.

4.8.6 Vibration.

4.8.6.1 Field vibration. Verify that the field vibration requirements of 3.8.6.1 have been met. A recommended method for verification may be found in 6.6.

4.8.6.2 Transportation vibration. Verify that the transportation vibration requirements of 3.8.6.2 have been met. A recommended method for verification may be found in 6.5.

4.8.7 Shock.

4.8.7.1 Field shock. Verify that the field shock requirements of 3.8.7.1 have been met. A recommended method for verification may be found in 6.5.

4.8.7.2 Transportation shock. Verify that the transportation shock requirements of 3.8.7.2 have been met. A recommended method for verification may be found in 6.5.

4.8.7.3 Bench handling. Verify that the bench handling requirements of 3.8.7.3 have been met. A recommended method for verification may be found in 6.5.

4.8.8 Sand and dust. Verify that the sand and dust requirements of 3.8.8 have been met. A recommended method for verification may be found in 6.5.

4.8.9 Salt and fog. Verify that the salt and flog requirements of 3.8.9 have been met. A recommended method for verification may be found in 6.5.

4.8.10 Explosive atmosphere. Verify that the explosive atmosphere requirements of 3.8.10 have been met. A recommended method for verification may be found in 6.5.

4.9 Electromagnetic interference (EMI). Verify that the EMI requirements of 3.9 have been met using the limits in Appendix A.

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4.10 Nuclear survivability. The MLRF shall be placed within the radiation volume of a radiated-wave, ground effects EMP simulator. The MLRF shall be subjected to ten (10) EMP pulses, five (5) with the mode switch in first RTN and five with the mode switch OFF. Three pulses of each of these sets shall be done with the MLRF orientated in the three orthogonal directions. The remaining two pulses of each of these sets shall be done with the MLRF mounted on the tripod and orientated in the two horizontal directions. Before and after each pulse, the MLRF shall be tested. This test shall consist of performing a successful range operation to a target.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Miniature Eyesafe Laser Infrared Observation Set (MELIOS), AN/PVS-6 is intended to provide the soldier with accurate range information to any selected target.

6.2 Acquisition requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification

.

b. First article tests required:

(1) Time frame required for submission of first article reports.

(2) Time frame required for approval of first article reports.

c. Sampling plans for Groups B, C and D testing. Unless otherwise specified sampling will be an accordance with ANSI/ASQC Z1.4.

d. Necessary actions required relative to Group C and Group D failures (see 4.4.3.2 and 4.4.4.2).

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e. Deliver all first article samples as specified by the procuring activity's contracting officer.

f. BA-6516/U (thionyl chloride) and BB-516/U (nickel cadmium) batteries may be obtained through the contracting officer.

g. The eyeshield and flash lamp stray emission, paragraphs 3.7.9.4 and 3.7.10, respectively, contain equipment which can be furnished by the procuring activity's contracting officer.

h. Issue of DoDISS to be cited in the solicitation and if required the specific issue of individual documents required.

i. Packaging requirements (see 5.1).

j. Environmental pollution prevention measures are contained in the packaging natural specifications or provided by the preparing activity for recommended disposability methods.

6.3 Definitions. See 3.12.

6.4 Subject term (key word) listing.

MELIOS
Eyesafe Laser
Range finder
MLRF

6.5 Environmental inspection procedures. The test methods and procedures of MIL-STD-781 and MIL-STD-810 are recommended, but are not mandatory for determining compliance with the environmental requirements of this specification.

6.5.1 Electromagnetic interference (EMI) tests. The MLRF will be tested for compliance with the requirements of MIL-STD-461, Class A3, Table 4-1 for laser devices, RE02 and RS03. The MLRF will be battery powered with the objective lens and laser window covered. During the radiated emissions tests, the MLRF will be fired at the sustained duty cycle rate while the receiver is recording the amplitude of the emissions. During the radiated susceptibility tests, the MLRF will be fired at the sustained duty cycle rate while observing the display. The radiated fields will not cause the MLRF to malfunction as observed by a range other than 0000.

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6.5.2 Reliability. The MLRF including battery or simulated battery mass, be tested in accordance with MIL-STD-781 using test plan 20-6 of MIL-HDBK-781.

6.5.3 Temperature.

6.5.3.1 High temperature-storage. The MLRF will be subjected to the tests of MIL-STD-810, Method 501.2, Procedure I. The objective lens will be protected during this test. During the tests the Hot Cycle with Induced Conditions as identified in Table 501.2-I will be used.

6.5.3.2 High temperature-operation. The MLRF will be subjected to the High temperature profile of Figure 6 with the following conditions. All temperature changes will be at 1.5°C/minute minimum. For these tests the temperatures of paragraph 3.8.1.1 will be used respectively. The profile will be conducted three (3) times. The operating temperature boresight shift test will be conducted in the third cycle. At the operating hot temperature the MLRF will be fired into burn paper or equivalent. At the conclusion of the tests the Unit Under Test will remain at Room Ambient Temperature until stabilized prior to post testing per 4.8.

6.5.3.3 Low temperature. The MLRF will be subjected to the tests of MIL-STD-810, Method 502.2, Procedures I, II, and III. For these tests the temperatures of 3.8.1.2 will be used respectively. Low temperature storage period will be a minimum of 24 hours. At the operating cold temperature the MLRF will be fired into burn film or equivalent. The operating temperature boresight shift test will be conducted during the operating cold temperature.

6.5.3.4 Temperature shock. The MLRF will be subjected to the test of MIL-STD-810, Method 503.2, Procedure I, in accordance with the temperatures of 3.8.1.3. During the performance of this test no diurnal cycling will be performed. Instead, the temperature chambers shall be maintained at the constant temperatures identified in 3.8.1.3 and the MLRF will remain in each chamber until it has stabilized.

6.5.4 Solar radiation. The MLRF will be subjected to the test of MIL-STD-810, Method 505.2, Procedure I, in accordance with the Hot-Dry Cycle identified herein (see 3.8.1.4). During the solar radiation test the MLRF will be fired during each diurnal cycle.

6.5.5 Immersion. The MLRF will be subjected to the test of MIL-STD-810, Method 512.2, Procedure I.

6.5.6 Humidity. The MLRF will be subjected to the test of MIL-STD-810, Method 507.2, Procedure I for ten (10) cycles of natural, cyclic high humidity in accordance with Table 507.2-I, Cycle 3.

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6.5.7 Altitude.

6.5.7.1 Storage and operation. The MLRF will be subjected to the test of MIL-STD-810, Method 500.2, Procedure II. In Step 2, the chamber pressure will be adjusted to an ambient air pressure equivalent to a 15,000 foot altitude, at an altitude rate of change between 1,500 ft/min and 2,000 ft/min. In Step 3, the MLRF laser will be fired a minimum of three times, and for each firing the laser emissions will be confirmed. Then the 15,000 foot equivalent altitude pressure will be maintained for 1 hour without MLRF operation. At the end of the 1 hour period the MLRF laser will be fired another three times, at a minimum, and laser emissions will be confirmed. At the completion of these firings proceed to Step 4. A complete inspection will be performed prior to the rapid decompression test.

6.5.7.2 Rapid decompression. The MLRF with battery and ancillary equipment, excluding the shipping and storage container, will be subjected to the test of MIL-STD-810, Method 500.2, Procedure III. In Step 1 the MLRF with battery installed, and all ancillary equipment will be packed in the laser rangefinder case as it would be when transported by an individual soldier. In Step 2, the equivalent altitude will be 15,000 feet. In Step 3, an equivalent altitude of 40,000 feet will be used.

6.5.8 Fungus. The MLRF, Laser Rangefinder case, tripod, and shipping and storage container will each be subjected to the test of MIL-STD-810, Method 508.3. The test duration will be 28 days.

6.5.9 Vibration.

6.5.9.1 Field vibration. The MLRF with a battery or dummy mass installed will be subjected to thirty minutes of random vibration conducted in accordance with MIL-STD-810, Method 514.3, Procedure I, using the limits of Figure 4 of this specification. The test item will be subjected to 10 minutes of vibration in each orthogonal axes, for a total time of thirty minutes.

6.5.9.2 Transportation vibration. The MELIOS will be subjected to the test of MIL-STD-810, Method 514.3, Procedure II.

6.5.10 Shock.

6.5.10.1 Field shock. The MLRF with a battery or dummy mass installed will be subjected to 3 shock pulses in each direction of each of the three (3) orthogonal axes, for a total of 18 shock pulses, using an a suitable shock machine. The MLRF will be securely mounted by the V block mounting point to a transponder and the shock pulses will be measured from a point on the MLRF as close to the mounting point as possible. The applied shock pulses will be of a terminal peak sawtooth shock pulse configuration as detailed in Figure 3 of this specification.

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6.5.10.2 Transportation shock. The MELIOS will be subjected to the tests of MIL-STD-810, Method 516.3, Procedure IV. The test height will be not less than 48 inches from the point of release to the point of impact. Evidence of damage or degradation of performance of the MLRF detected during the test or upon post test inspection will constitute failure of this test. Damage to the MELIOS shipping and storage container will be recorded and reported in the test report, but will not be considered as a failure unless such damage renders the shipping case ineffective in protecting the MLRF, and/or the shipping and storage container is damaged in such a manner as to preclude removal of the MLRF by one person without the use of tools.

6.5.10.3 Bench handling. The MLRF will be subjected to the test of MIL-STD-810, Method 516.3, Procedure VI.

6.5.11 Sand and dust. The MLRF will be subjected to the tests of MIL-STD-810, Method 510.2 Procedures I and II. These tests will be performed with all lens caps installed. For Procedure I, either red chine clay or silica flour will be used, an air velocity of 8.9 m/s (1750 ft/min) will be used in Steps 2 and 7, and a temperature of +49°C (+120°F), will be used in Step 5. For procedure II, the temperature will be +49°C (120°F), the air velocity will be 18 to 29 m/s (3540 to 5700 ft/min), and a sand concentration of 2.2 ± 0.5 g/m3 (0.0623 ± 0.015 g/ft3) will be used. The applicable duration of exposure for Step 4 will be 90 minutes per face of the MLRF. The MLRF will not be operated during performance of Procedures I and II. For these tests only, degraded finish (excluding covered optical surfaces) will be permissible, with the exception that operational and safety markings will not be obscured to the extend of illegibility.

6.5.12 Salt fog. The MLRF and will be subjected to the test of MIL-STD-810, Method 5112.2, Procedure I.

6.5.13 Explosive atmosphere. The MLRF will be subjected to the test of MIL-STD-810, Method 511.2, Procedure I. In Step 3 the ambient temperature will be +49°C (120°F), and in Step 4 the test altitude will be 15,000 feet. The MLRF will be configured for normal operation, and the atmosphere of the test chamber will contain fuel weight calculated to be 3.8 percent by volume of the test atmosphere. The MLRF, when operated as specified in Step 7, will not ignite the atmosphere in the test chamber. Provisions will be made for monitoring operation of the MLRF while in the test chamber.

6.6 Safety. Identification of potential hazardous areas within the MLRF laser device can be found in MIL-STD-882. MIL-STD-882 is to be used only as a guide.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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APPENDIX A

ELECTROMAGNETIC INTERFERENCE (EMI) VERIFICATION PROGRAM FOR THE LASER INFRARED OBSERVATION SET, AN/PVS-6

A.1. SCOPE

A.1.1 Scope. This appendix outlines and specifies the requirements for the electromagnetic interference (EMI) verification program applicable to Miniature Eyesafe Laser Infrared Observation Set (MELIOS). It is a mandatory part of the specification, and is intended for compliance. The MELIOS Laser Rangefinder, herein will be referred to as the MLRF.

A.1.2 The EMI verification program for the MLRF shall be completed as per the following test plans:

a. EMI Qualification Test Plan procedures are to be performed on first article equipments as specified in the contract or purchase order.

b. EMI Production Verification Test Plan procedures are to be performed on samples of production equipments as specified in the contract or purchase order.

A.2. APPLICABLE DOCUMENTS

Not applicable to this appendix.

A.3. TEST CONDITIONS

A.3.1 The equipment shall be set up in a shielded enclosure with the test equipment.

A.3.2 The test shall be conducted at normal room ambient temperature of 20° C to 30°C (68°F to 86°F).

A.3.3 During radiated emission tests, the MLRF shall be fired at the sustained duty cycle rate while the receiver is recording the amplitude of the emissions. During the radiated susceptibility tests, the MLRF shall be fired at the sustained duty cycle rate while observing the display. The radiated fields shall not cause the MLRF to malfunction as observed by a range other than 0000. Any range readout other than 0000 or MLRF emissions greater than the limits set herein shall be considered a failure.

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A.4. VERIFICATION PROCEDURE

A.4.1 Radiated emissions, electric field, 10 kHz to 18 GHz. The equipment is to be tested for radiated emissions of the electric field over the frequency range of 10 kHz to 18 GHz using the measurement setup of Figure A-1. The emissions shall not exceed the levels shown in Figure A-2. Above 30 MHz, the limits shall be met for both horizontally and vertically polarized fields.(RE102)

A.4.2 Radiated susceptibility, electric field, 10 KHz to 40 GHz. The equipment is to be tested for radiated susceptibility to the electric field over the frequency range of 10 KHz to 40 GHz using the measurement setup of Figure A-3. The equipment shall not exhibit any malfunction, degradation of performance, or deviation from specified indications, beyond the tolerances indicated in this specification, when subjected to the electric fields shown below in Table A-I. Up to 30 MHz, the limits shall be met for vertically polarized fields. Above 30 MHz, the limits shall be met for both horizontally and vertically polarized fields. Circular polarized fields are not acceptable. (RS103)

Table A-I. Radiated susceptibility limits - electric field

Frequency range	Limit level (volts/meter)
10 kHz-2 MHz	20
2 MHz - 30 MHz	50
30 MHz - 1 GHz	50
1 GHz - 18 GHz	50
18 GHz - 40 GHz	50

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(RE102)

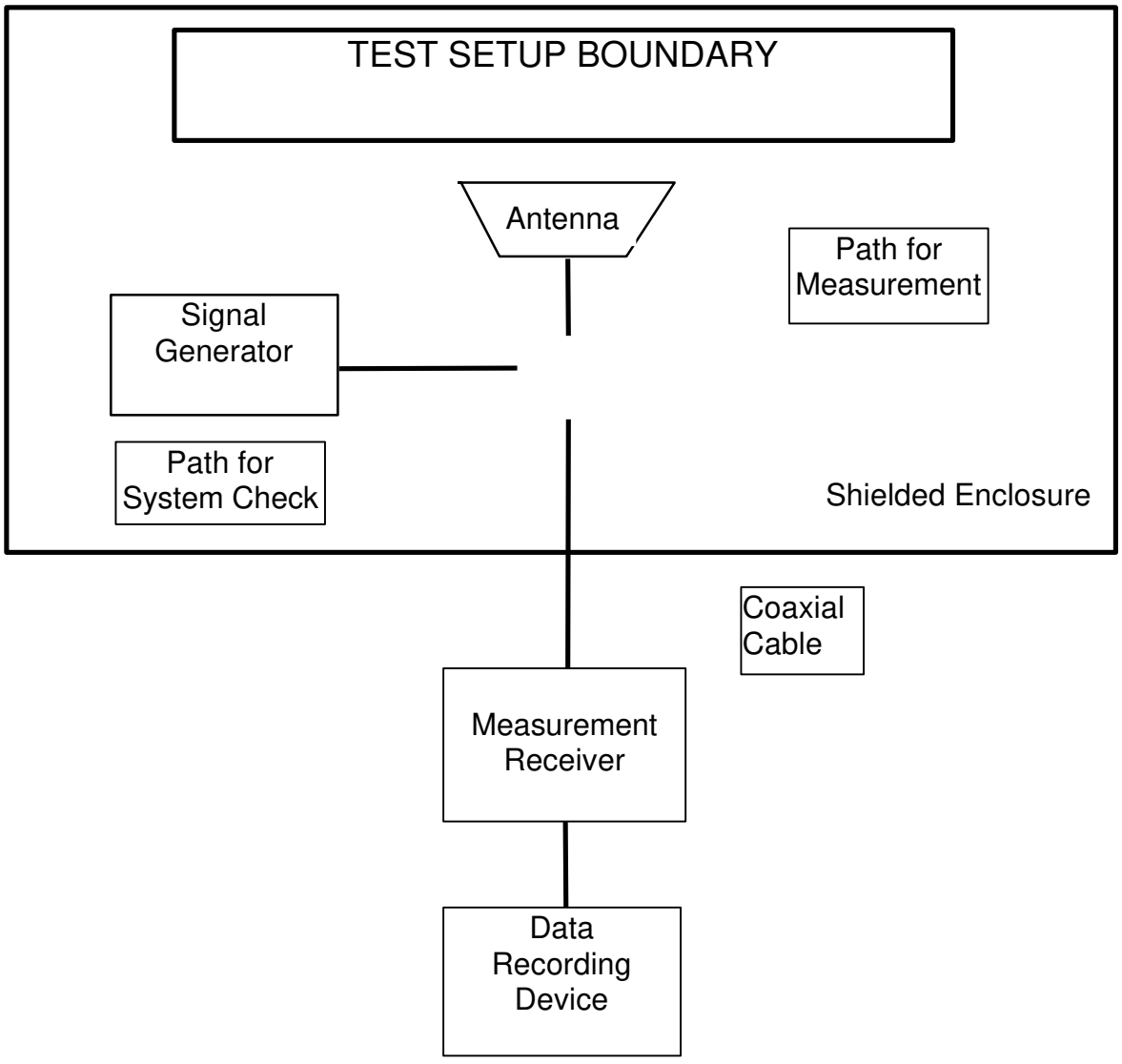


FIGURE A-13. Test setup, radiated emissions, electric field, 2 MHz to 18 GHz.

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(RE102)

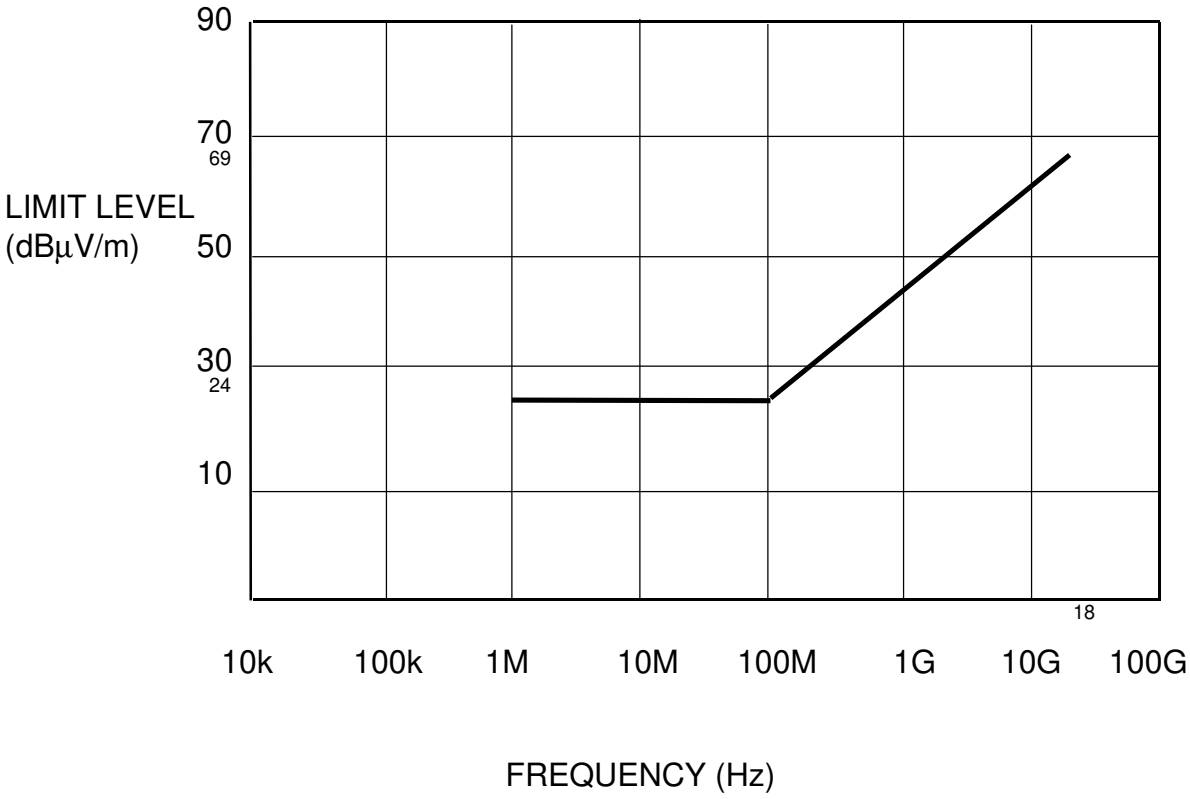


FIGURE A-14. Radiated Emission limit, electric field

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(RS103)

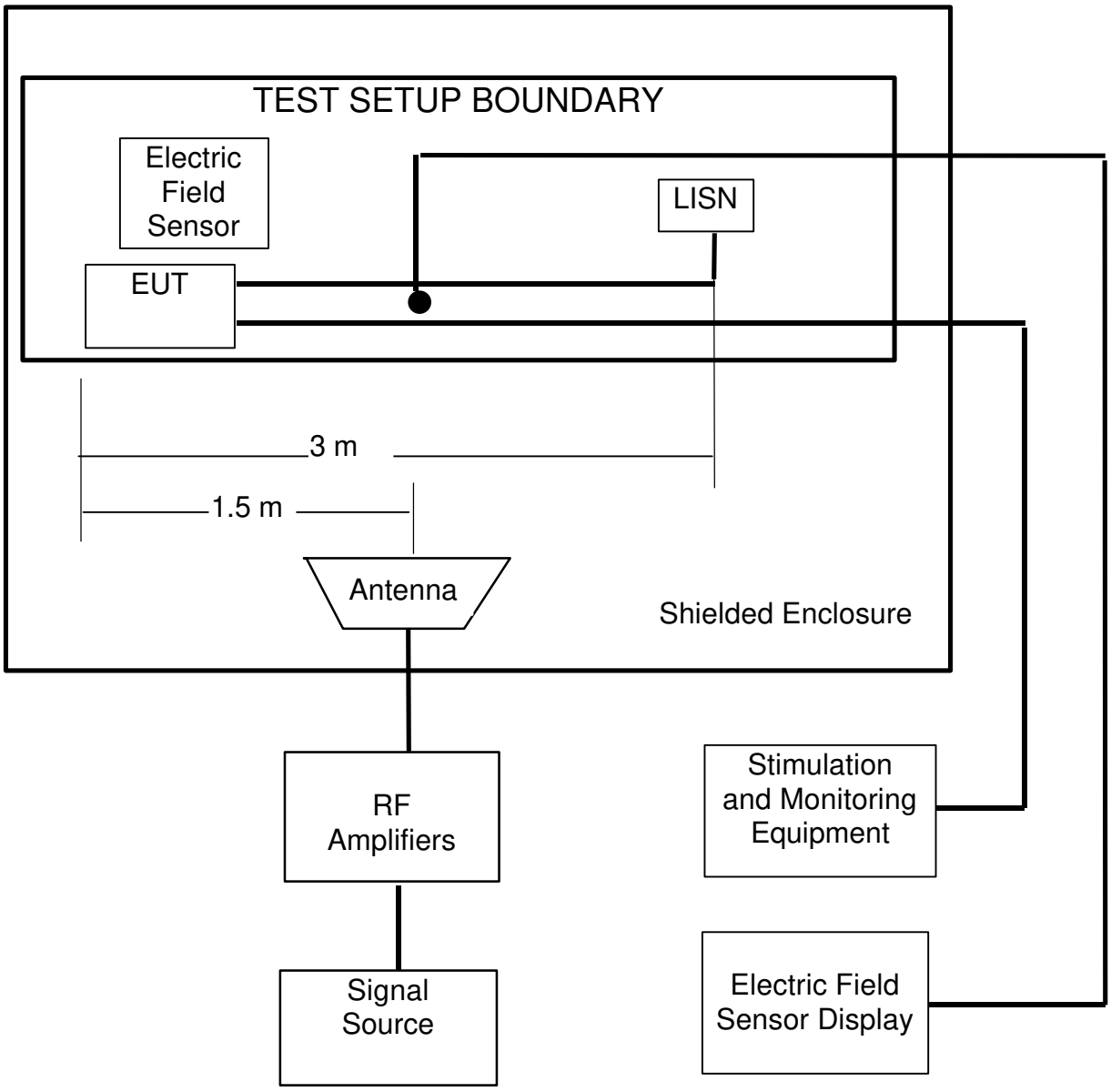


FIGURE A-17. Test setup, radiated susceptibility, electric field, 2 Mhz to 40 GHz.

(RS103)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-49320A	**2. DOCUMENT DATE (YYMMDD)** 97/07/03

3. DOCUMENT TITLE LASER-INFRARED OBSERVATION SET, AN/PVS-6

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME US Army Communications-Electronics Command	b. TELEPHONE (Include Area Code) (1) Commercial (732) 532-9913 (2) AUTOVON 992-9913
c. ADDRESS (Include Zip Code) ATTN: AMSEL-LC-LEO-E-EP Fort Monmouth, NJ 07703-5023	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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